

Energy Sources

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Introduction :-

Energy is an important aspect of human activity. In the early days the energy demands were met primarily by muscular effort, wind and water currents, fuel-wood, direct solar warming and such other sources. With the progress of industrial civilization, these sources became less important and fossil fuels such as coal, lignite, petroleum and natural gas began to play a vital role as sources of energy.

In fact, energy consumption today is related to the extent of industrialization of the country and the standard of living of its people. There has been a ten-fold growth in energy consumption over the past three decades and it has been found to increase thirty times by 2050 which is a challenging proposition to be tackled.

The energy sources can be broadly classified into two types - namely,

- (1) fuels - eg:- coal, petroleum, natural gas, peat etc
- (2) nonfuels - eg:- hydro, solar, wind, geothermal energy.

Fuels :-

Definition :- A chemical substance that produces light and heat energy when burnt in excess of air & oxygen is called chemical fuels.

They invariably contain carbon and hydrogen as major elements. When chemical fuels undergo combustion, carbon and hydrogen are converted into carbon dioxide and water vapour respectively. Since the products of combustion have less heat content

than the reactants, chemical fuels release heat energy during combustion process.

Chemical fuels are primarily used as a source of heat and power. The important application of chemical fuels are as reducing agents in metallurgical operations and to produce electrical and mechanical energy.

Classification of fuels :-

The fuels have been classified according to their (1) occurrence and (2) the state of aggregation.

According to the occurrence -

(i) Natural & primary fuels → which are found in nature as such.

eg:- wood, peat, coal, petroleum, Natural gas etc.

(ii) Secondary & Artificial fuels → are those which are prepared from the primary fuels.

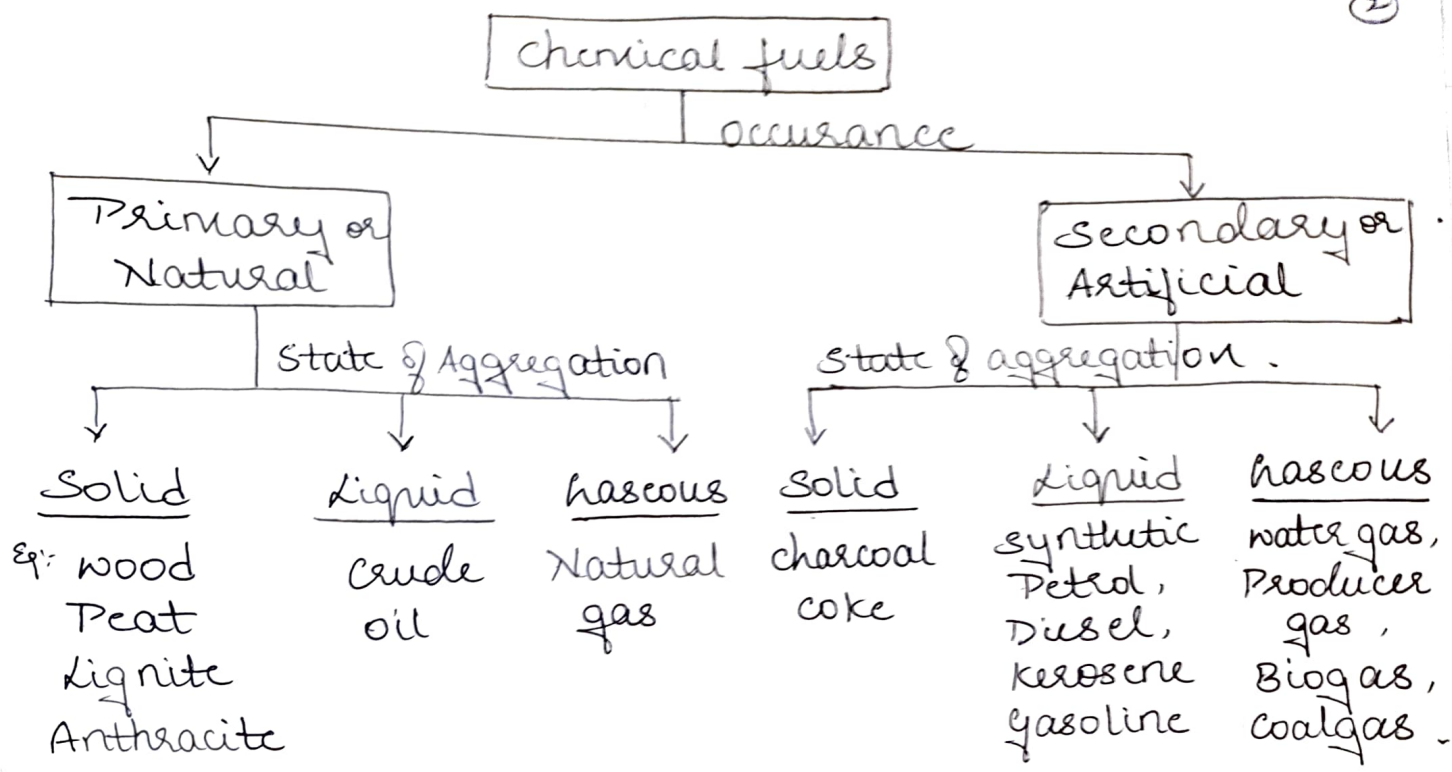
eg:- charcoal, coke, kerosene oil, Diesel oil, petrol, oil gas, producer gas etc.

The second classification is based upon their state of aggregation like -

(a) solid fuels

(b) liquid fuels

(c) hazy fuels.



Calorific value :-

The most important property of the fuel to be taken into account is its calorific value @ the capacity to supply heat.

The calorific value of a fuel can be defined as "the total quantity of heat liberated when a unit mass (or volume) of the fuel is burnt completely."

units of Heat :-

1) calorie :- is the amount of heat required to raise the temperature of one gram of water through one degree centigrade. (15-16°C).

2) Kilocalorie @ kilogram centigrade unit :- is equal to 1000 calories. This is the unit of metric system and may be defined as the quantity of heat required to raise the temperature of one kilogram of water through one degree centigrade. 1 Kcal = 1,000 cal.

3) British thermal unit :- is defined as the quantity of heat required to raise the temperature of one pound of water through one degree Fahrenheit ($60-61^{\circ}\text{F}$).

$$1 \text{ B.Th.U} = 252 \text{ cal} = 0.252 \text{ Kcal}.$$

$$1 \text{ Kcal} = 3.968 \text{ B.Th.U}.$$

4) Centigrade heat unit (C.H.U) :- is the quantity of heat required to raise the temperature of 1 pound of water through one degree centigrade.

$$1 \text{ Kcal} = 3.968 \text{ B.Th.U} = 2.2 \text{ C.H.U}.$$

The SI unit of heat is KJ/kg .

Gross calorific value (G.C.V) :-

It is defined as the amount of heat liberated when unit quantity of fuel is completely burnt in air and the products of combustion are cooled to room temperature.

Normally fuels contain carbon and hydrogen and these on combustion produce carbon dioxide and water vapour respectively. On cooling these products water vapour condenses to water and liberates latent heat of steam. G.C.V includes latent heat of steam and therefore has higher value.

Thus it is also called as Higher calorific value i.e., H.C.V.

Net calorific value (NCV) :-

It is defined as the amount of heat liberated when unit quantity of fuel is completely burnt in air and the products of combustion are not cooled to room temperature.

In actual combustion practice, the combustion products are not allowed to cool but are let off into the atmosphere. The net calorific value is therefore always lower than gross calorific value and represents the heating capacity of fuel.

Thus it is also called as low calorific value LCV.

$$\text{Net Calorific value} = \text{Gross calorific value} - \text{Latent heat of steam.}$$

Characteristics of a good fuel :-

The performance of a fuel largely depends on its calorific value and how cleanly it burns. The following are the desirable properties of a good chemical fuel.

1) High calorific value :-

Calorific value is the direct measure of the efficiency of a fuel. Higher the calorific value, more efficient is the fuel. Thus a good fuel must have high calorific value.

2) Moderate Ignition temperature :-

Ignition temperature is the minimum temp to which the fuel is to be heated to start burning.

6) Combustion control :-

One can avoid a large wastage of valuable fuel if its combustion rate can be properly regulated and burning can be stopped immediately when desired.

7) Ease of availability :-

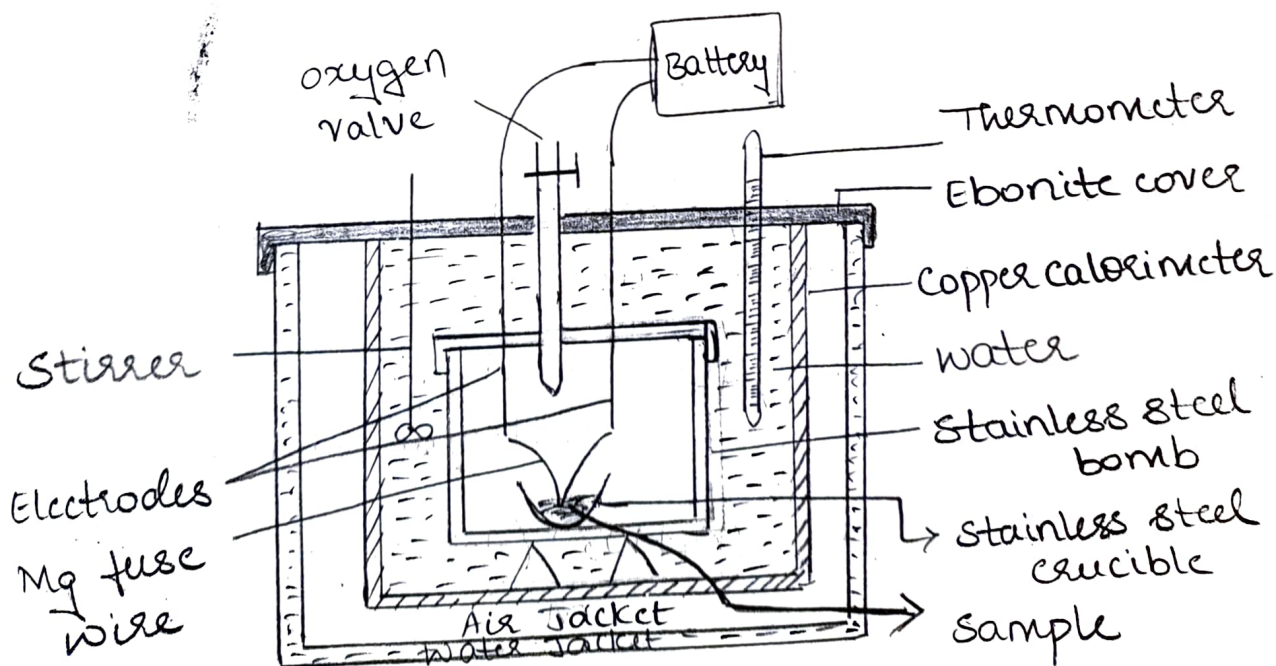
Fuel must be readily available in abundant amount and its cost must be minimum.

8) Ease of storage and transport :-

Fuel should not demand more space and sophisticated arrangements for their storage and transportation and there must be minimum risks of fire hazards during their transportation & storage.

Bomb Calorimeter - Determination of calorific value

Calorific value of a solid or liquid fuel is determined by means of a bomb calorimeter.



A known weight of the sample is placed in a crucible within a sealed vessel called bomb. It has an inlet ~~with~~^{valve} for adding oxygen under pressure and also an electric ignition coil. After the sample has been placed in the bomb, the bomb is sealed and filled with oxygen. Next to it placed is a large insulated container i.e., calorimeter filled with accurately measured quantity of water. The calorimeter is equipped with a stirrer and a thermometer. The combustion of fuel is initiated by passing electric current through the coil in contact with the sample. The sample ignites and undergoes combustion. The temperature of the water before and after combustion is carefully measured. The heat lost by burning sample is the heat gained by the surrounding water and calorimeter.

observations and calculations :-

mass of the fuel = m g

mass of water in calorimeter = w_1 g

mass of water equivalent of calorimeter = w_2 g

Initial temperature of water = $t_1^\circ\text{C}$

Final temperature of water = $t_2^\circ\text{C}$

Heat lost by ' m ' gm of fuel = Heat gained by water +
Heat gained by calorimeter.

$$\text{Gross Calorific value} = \frac{(w_1 + w_2)(t_2 - t_1)}{m} \times \frac{\text{calories/kg.}}{4.187 \text{ KJ/kg}}$$

specific heat of water = 4.187 KJ/kg .

To calculate NCV, we must know the % of hydrogen in the fuel. Let the % of hydrogen be x . (5)



$$2\text{kg} \longrightarrow 18\text{kg}$$

$$x\% \longrightarrow ? \quad \frac{x}{100} \times \frac{18}{2} = \frac{0.09x}{100} = 0.09x$$

$$\text{Latent heat of steam} = 587 \text{ cal/g}$$

$$\therefore \boxed{\text{NCV} = \text{GCV} - 0.09x \times 587 \times 4.187 \text{ kJ/kg}}$$

Problems :-

1) A sample of coal contains 95% C, 3% ash and 2% hydrogen. When 0.95g of this coal sample when subjected to combustion in a bomb calorimeter raised the temperature of 2800g of water from 25°C to 28.5°C. Calculate GCV and NCV of the coal sample. Given that water equivalent of calorimeter is 450g and latent heat of steam is 587 cal/g.

Solution :-

$$\text{GCV} = \frac{(w_1 + w_2)(t_2 - t_1)}{m} \times 4.187$$

$$= \frac{(2800 + 450)(28.5 - 25)}{0.95} \times 4.187 = \frac{3250 \times 3.5}{0.95} \times 4.187$$

$$= \underline{\underline{50133.81 \text{ kJ/kg}}}$$

$$\text{NCV} = \text{GCV} - 0.09H \times 587 \times 4.187$$

$$= 50133.81 - 0.09 \times 2 \times 587 \times 4.187$$

$$= 50133.81 - 442.398$$

$$= \underline{\underline{49691.4 \text{ kJ/kg}}}$$

2) A sample of coal contains 94% carbon, 3.5% ash and 2.5% hydrogen. When 0.93g of this coal sample when subjected to combustion in a bomb calorimeter raised the temperature of 2650g of water from 26.8°C to 28.5°C . Calculate the HCV and NCV of the coal sample. Given that the water equivalent of calorimeter is 430g, latent heat of steam is 587 cal/g and specific heat of water is 4.187 kJ/kg $\leftarrow$ just

Solution:-

552.998

$$\text{HCV} = 23573.26 \text{ kJ/kg}$$

$$\text{NCV} = 23020.26 \text{ kJ/kg}.$$

Petroleum:-

Petroleum is the major source of liquid fuel. The crude petroleum is a mixture of solid, liquid and gaseous hydrocarbons formed by the fossil remains of the plants and animals.

The average composition of crude petroleum is
 $\text{C} = 79.5 \text{ to } 87.1\%$, $\text{H} = 11.5 \text{ to } 14.8\%$, $\text{S} = 0.1 \text{ to } 3.5\%$,
 $\text{N} + \text{O} = 0.1 \text{ to } 0.5\%$.

Mining of Petroleum:-

Petroleum is brought to the surface of the earth by drilling wells. Crude petroleum reaches the surface of the earth as a foam consisting of a mixture of solid, liquid and gaseous hydrocarbons, sand and water in suspension. Petroleum obtained from the oil wells is known as crude oil.